

# CS & IT ENGINEERING



## Algorithms

## Greedy Algorithms

Lecture No.- 06



By- Aditya sir

# Recap of Previous Lecture



Topic

Topic

Intro to MCST



# Topics to be Covered



Topic

Topic

Topic

MCST

PYQs





## About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on LinkedIn where I share my insights and guide students and professionals.





Telegram Link for Aditya Jain sir: [https://t.me/AdityaSir\\_PW](https://t.me/AdityaSir_PW)

TTTe



## Topic : Applications (real-life) of spanning Trees

### Networking:

- 1) Broadcasting
- 2) Multicasting
- 3) Electronics: Circuits

MCST → Optimization Problem → Unique min cost



## Topic : Applications (real-life) of spanning Trees

Algorithms for Construction of Minimum Cost Spanning Trees:

- 1) Prim's Algorithm
- 2) Kruskal Algorithm
- 3) Dijkstra's Algorithm → Question

Note: different from Dijkstra's Single Source Shortest Path Algo.



SSSP



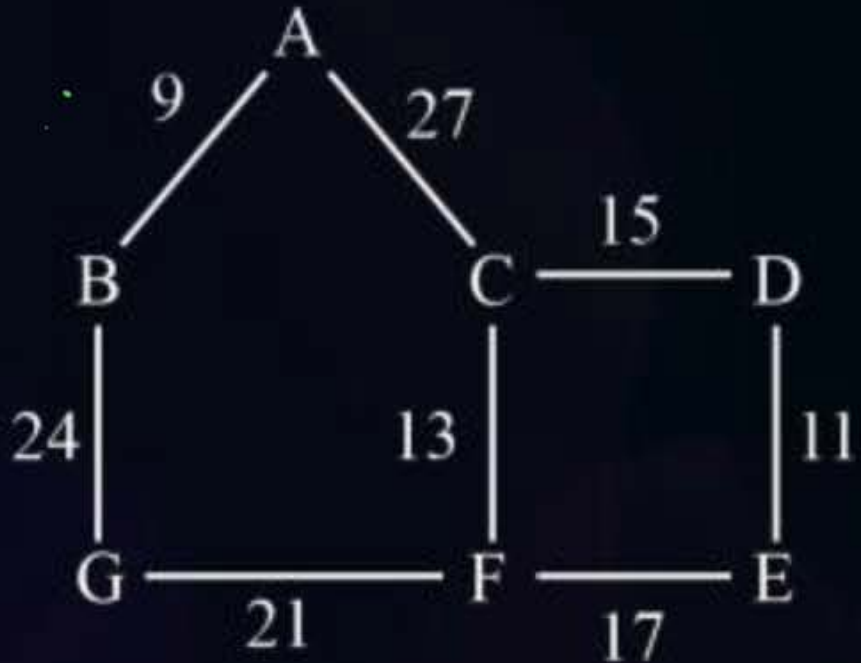


## Topic : Applications (real-life) of spanning Trees

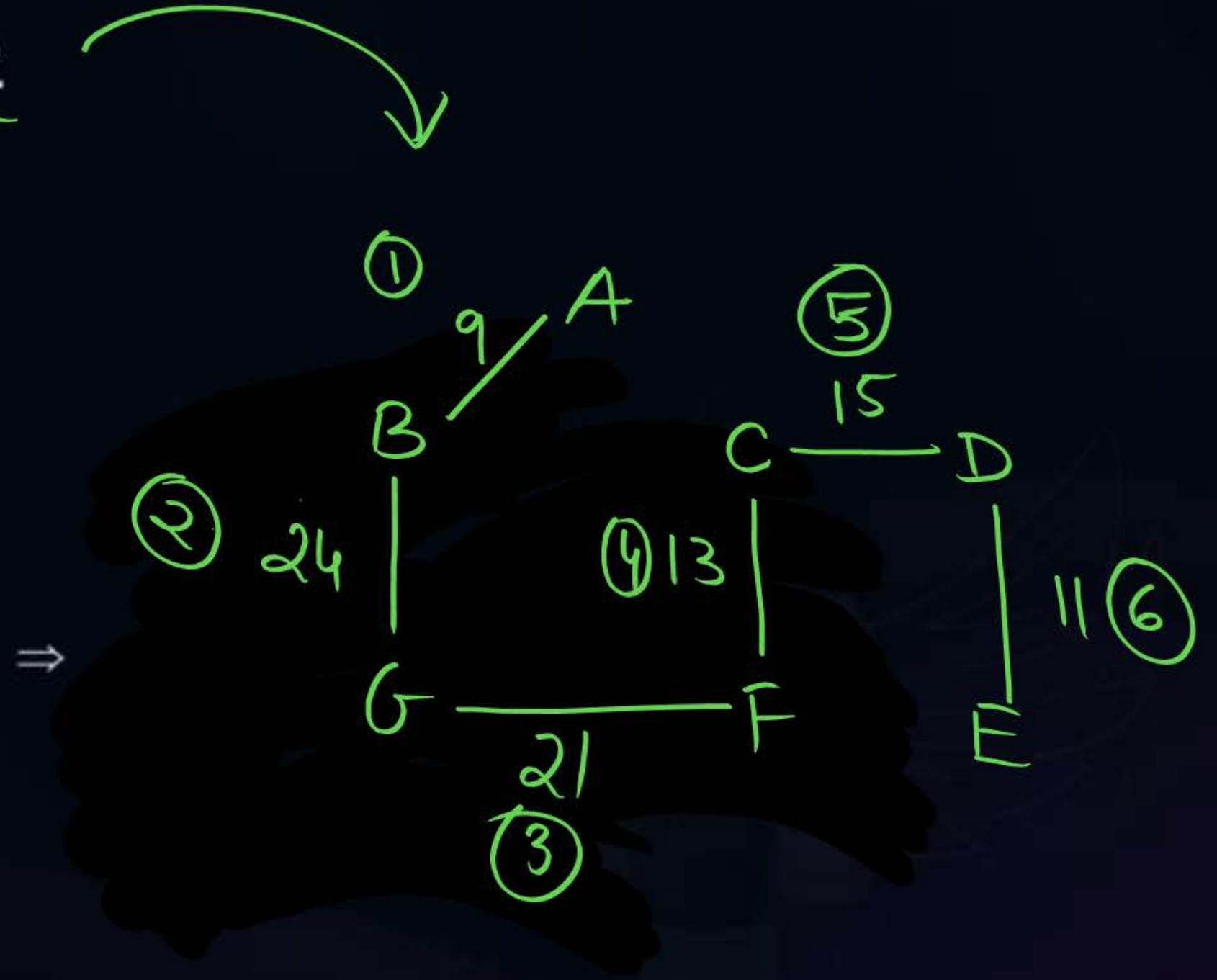
### Prim's Algorithm: Construction of MCST.

Given Graph,  $G = (v, e)$

$v \rightarrow n, e \rightarrow (n - 1) \rightarrow$  MCST



$$\text{Cost} = 9 + 24 + 21 + 13 + 15 + 11 = 93$$



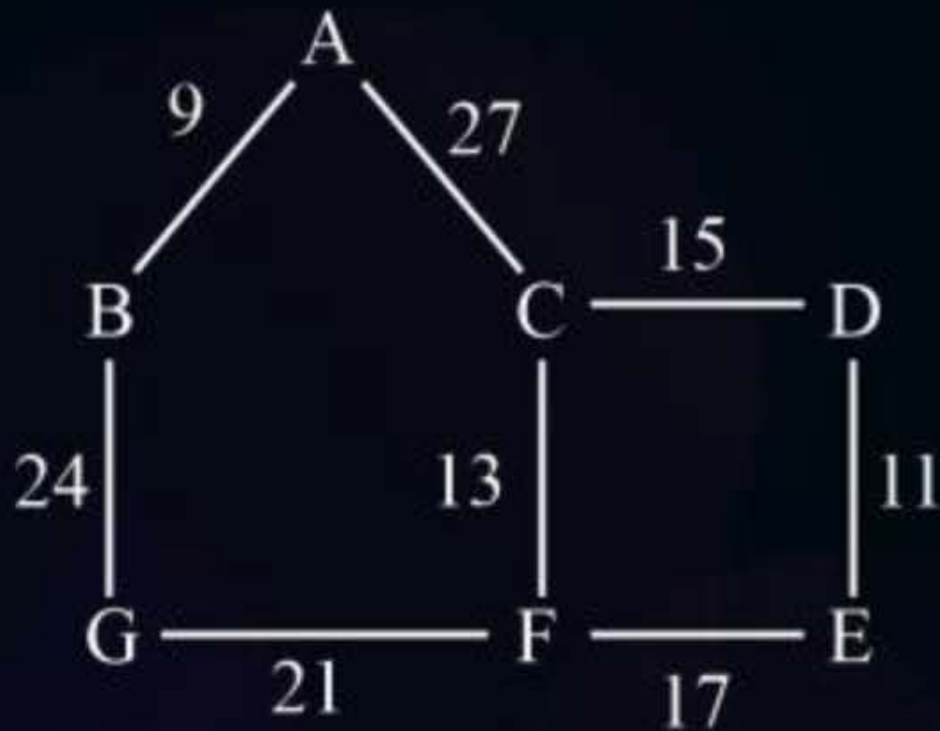




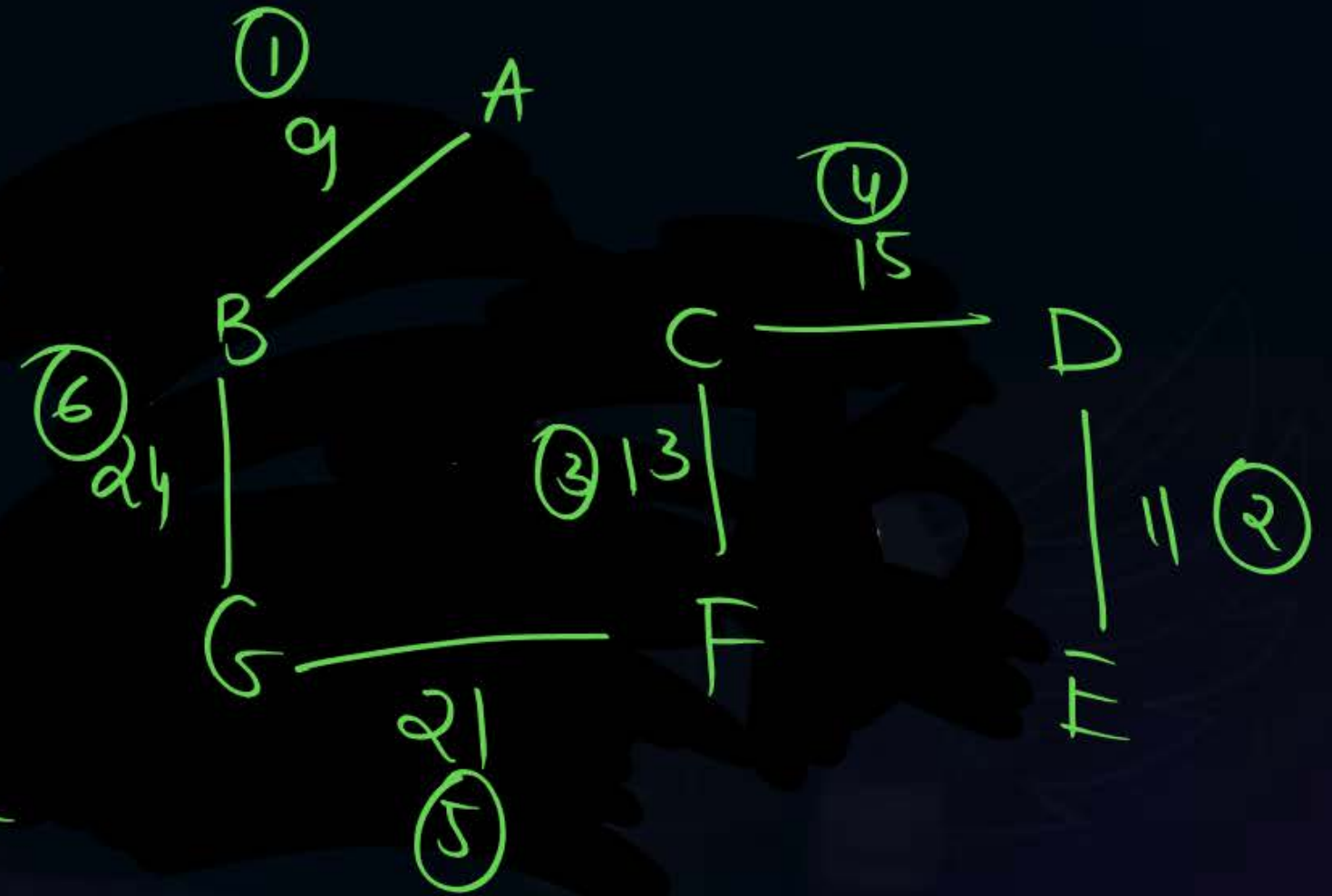
## Topic : Applications (real-life) of spanning Trees

**Kruskal Algorithm:** Min Heap + Set

[9, 11, 13, 15, 17, 21, 24, 27]



$\Rightarrow$



$$\text{Cost} = 9 + 24 + 21 + 13 + 15 + 11 = 93$$



## Topic : Greedy Algorithm



For a given Graph :  $G(V, E)$ ,  $|V| = n$ ,  $|E| = e$

- 1) Prims Algo:  $TC = O(n^2)$
- 2) Kruskal Algo:  $TC = O(e \log e)$
- 3) Prims Algo always maintains Tree Structure property at each step, whereas Kruskal may or may not.
- 4) Cost of Spanning Tree (MCST) by both the approaches will always be same.
- 5) The tree structure of the MCST obtained by both may or may not be same.





## Topic : Greedy Algorithm

- a) <sup>in Graph</sup> If all the edges have unique/distinct cost, then the structure of MCST by both algo is also same.
- b) If there are duplicate weighted edges, then the structure of MCST may or may not be the same.

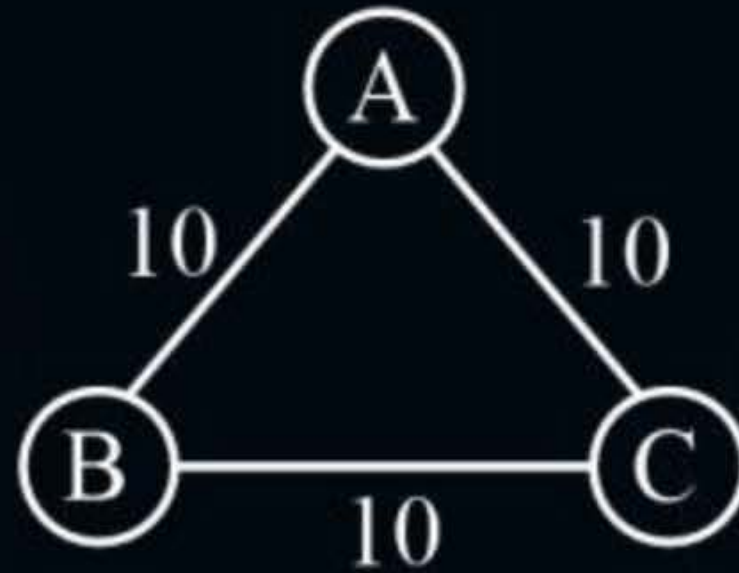
v. imp



## Topic : Greedy Algorithm



**Example:**  $n = 3, e = 3$



Cost (MCST) = 20

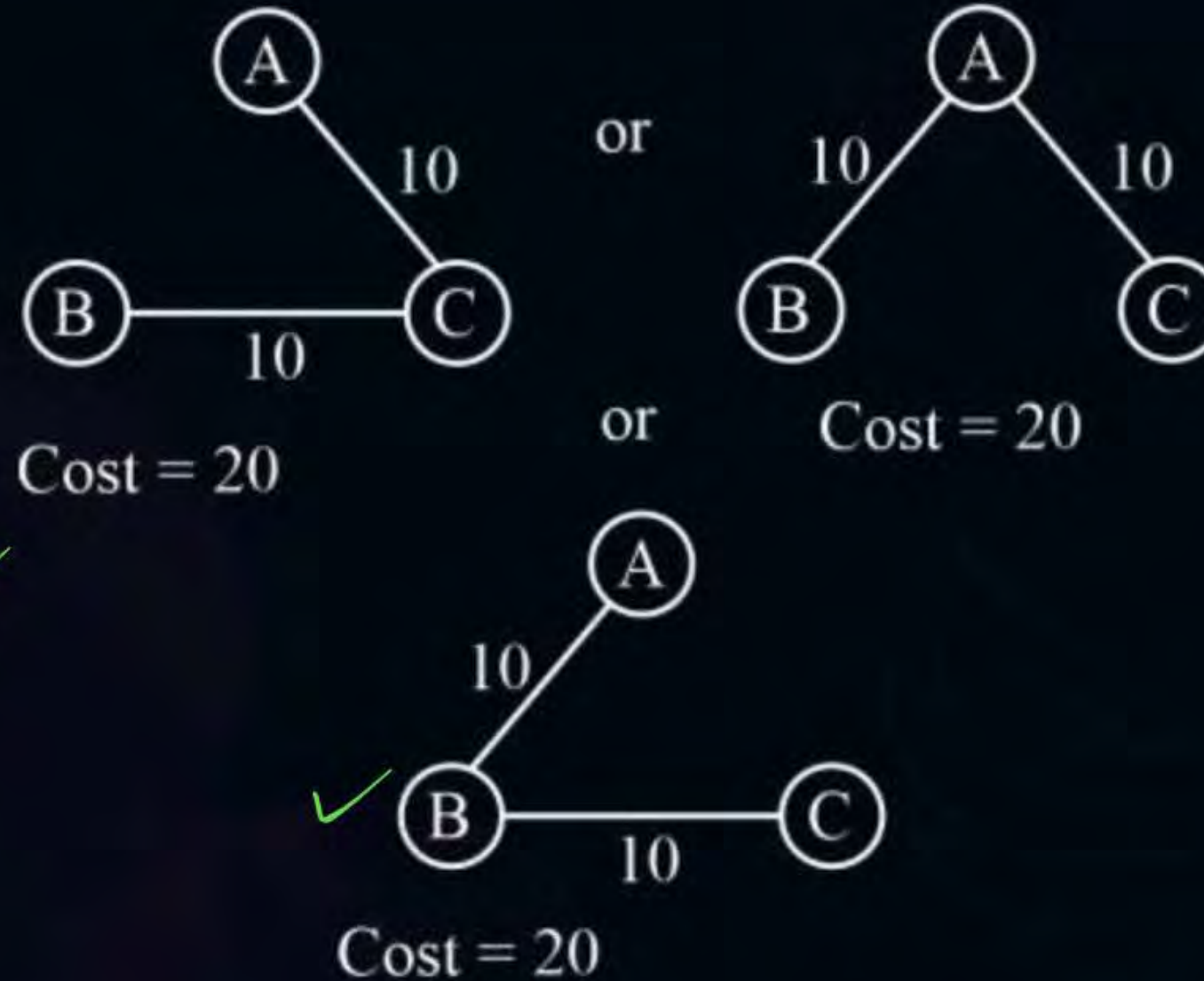




## Topic : Greedy Algorithm



### MCST:



- Hence, cost of MCST is always unique. But, there can be multiple MCSTs (as there are duplicate weighted edges)



## Topic : Greedy Algorithm



### Important:

#Q. If all the edges in the  $G(V, E)$  are equal (equi-weighted graph), then the cost of MCST = ?

$$|V| = n$$

$$\underline{\text{cost} = c}$$

$$\text{Cost of MCST} = \underline{c * (n-1)}$$

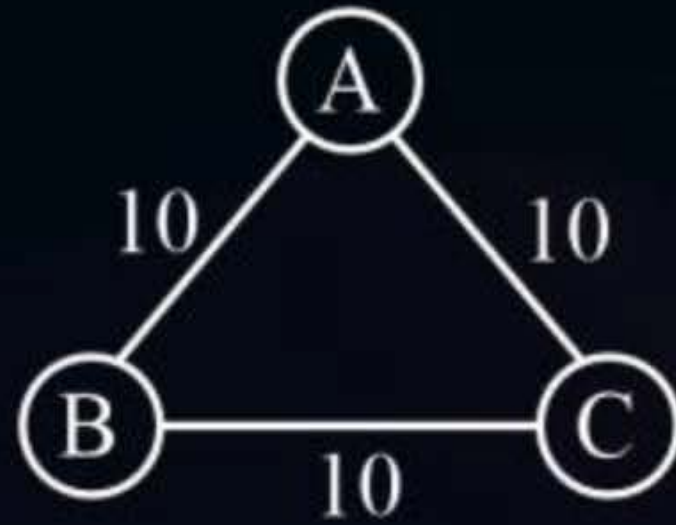




## Topic : Greedy Algorithm

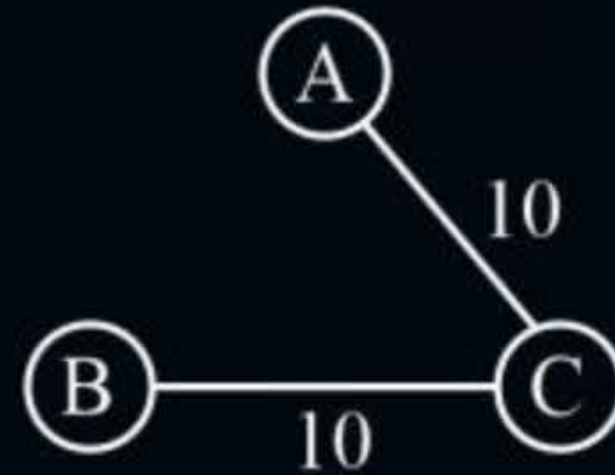


10, 10, 10



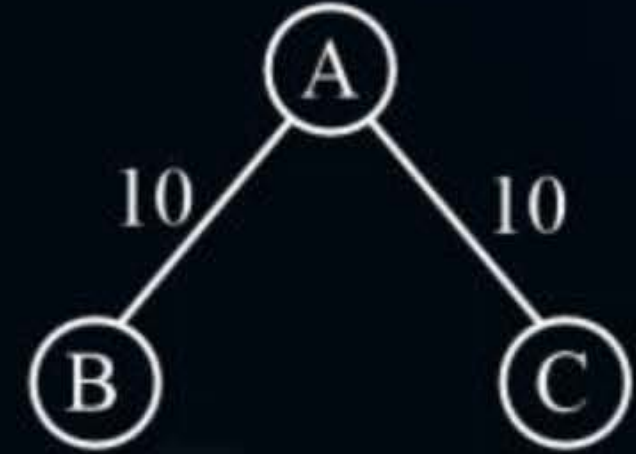
$$C = 10$$

$$n = 3$$



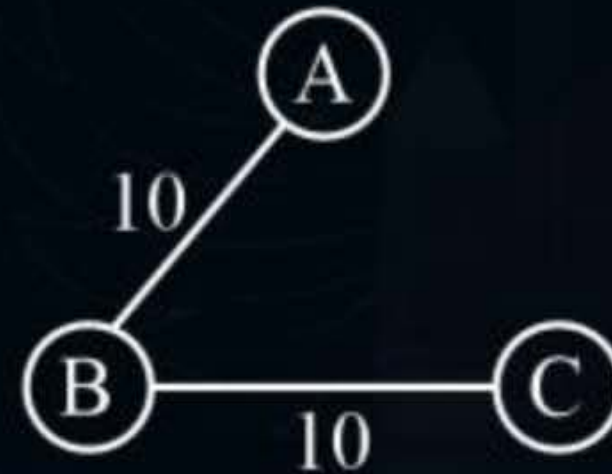
Cost = 20

or



Cost = 20

or



Cost = 20



## Topic : Greedy Algorithm



### Dijkstra's MCST Construction Algorithm:

Steps:

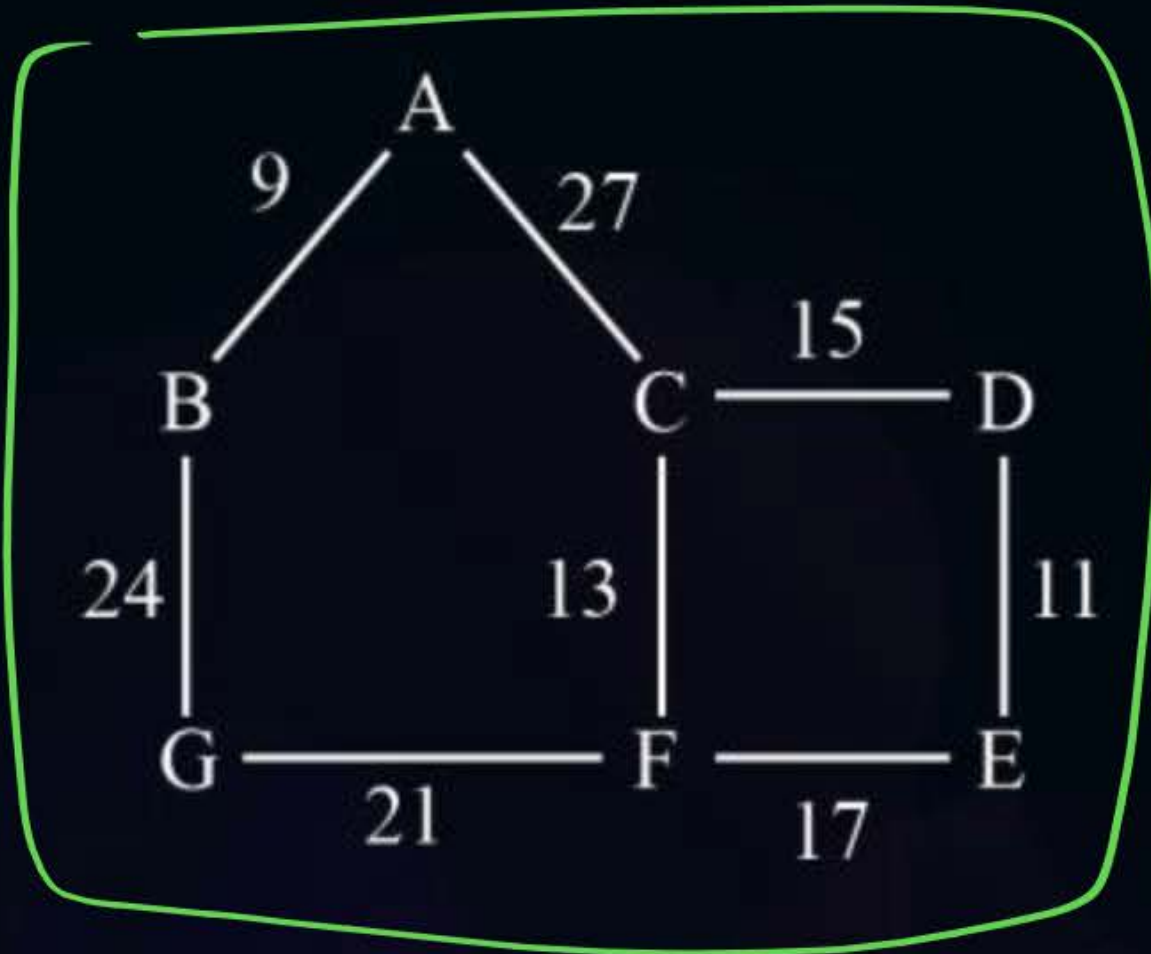
- 1) Randomly Pick edges (consider all of them)
- 2) Only when cycle gets formed, remove the max cost edge from that cycle and continue until all edges are considered.





## Topic : Greedy Algorithm

### Dijkstra's MCST Construction Algorithm:



$\Rightarrow$



~~MCST Cost = 9 + 24 + 21 + 13 + 15 + 17 = 99~~

Possible mistake as stopped after  $(n - 1)$  edges

Actual MCST Cost = 93

[NAT]

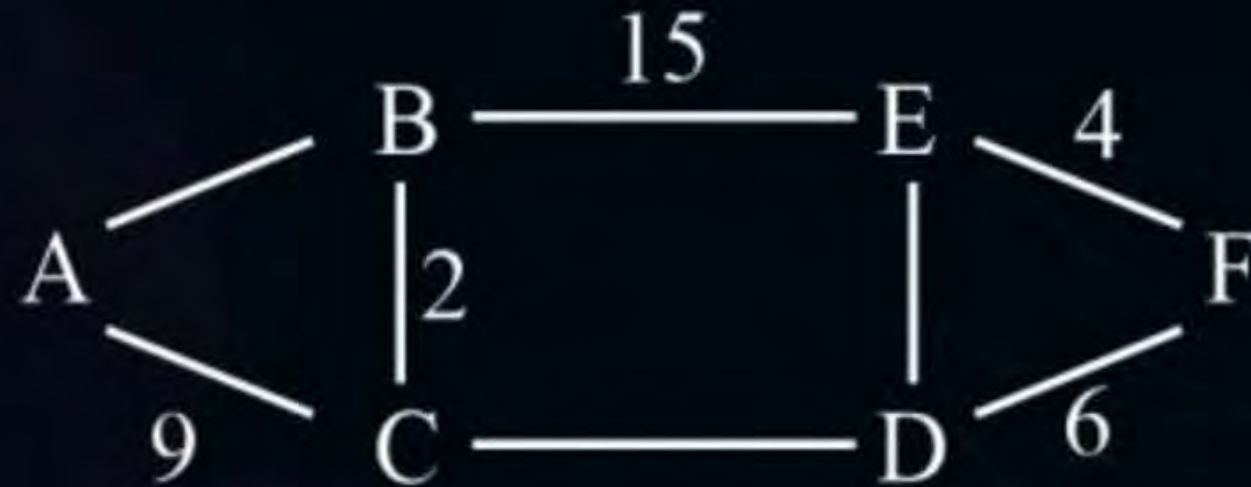


PYQ

#Q. Consider the following graph whose minimum cost spanning tree marked with edges values have a weight of 36.

Minimum possible cost of all the edges of the graph G is \_\_\_\_\_?

Assume that all the edges have distinct cost.



69

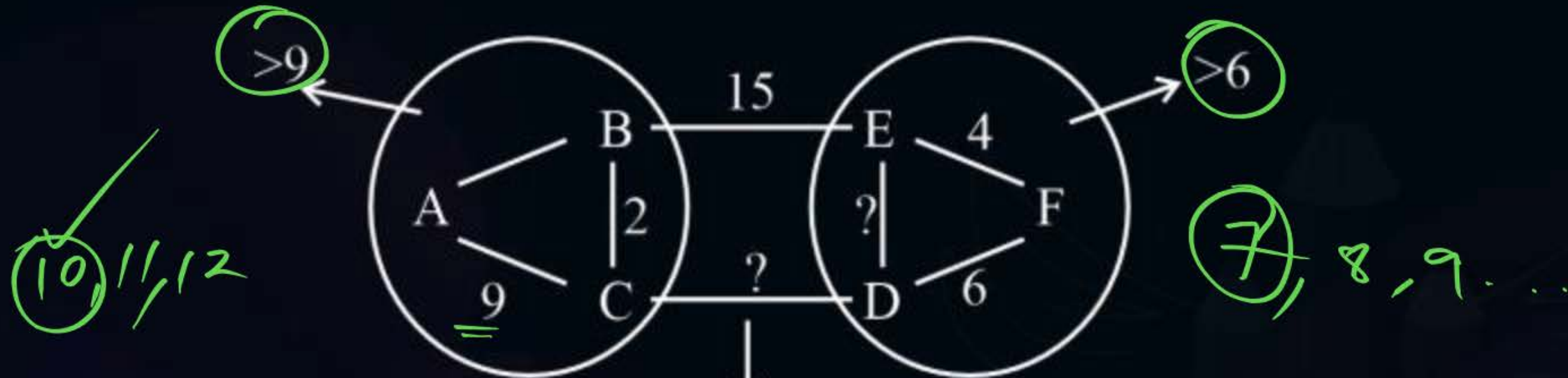




## Topic : Greedy Algorithm



As per Dijkstra MCST Approach:



$$\begin{aligned} 36 + 10 &= 46 \\ + 16 &= 62 \\ + 7 &= 69 \end{aligned}$$

16, 17, 18...

[NAT]

PYQ



#Q. Let  $G$  be a complete undirected graph with 4 vertices and edge weights are  $\{1, 2, 3, 4, 5, 6\}$ . The maximum possible weight that a minimum weight spanning tree can have is \_\_\_\_\_.

9.5%

max

MCST<sub>1</sub>

MCST<sub>2</sub>

MCST<sub>k</sub>

6?



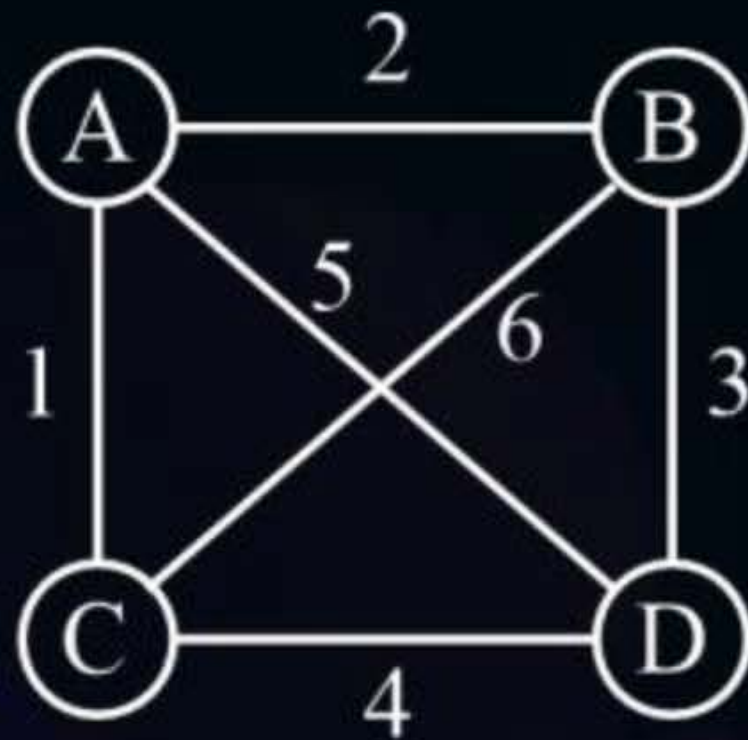
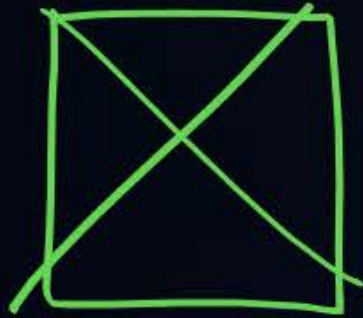


## Topic : Greedy Algorithm



**Solution:**

**G1:**



MCST  $\rightarrow$



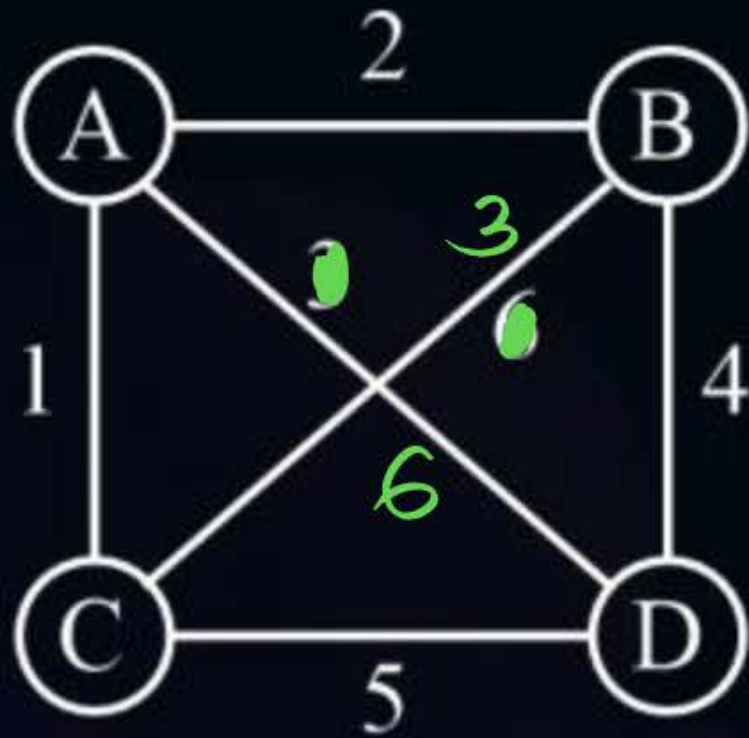
$$\text{Cost} = 1 + 2 + 3 = 6$$



## Topic : Greedy Algorithm



G2:



MCST →



$$\text{Cost} = 1 + 2 + 4 = 7$$

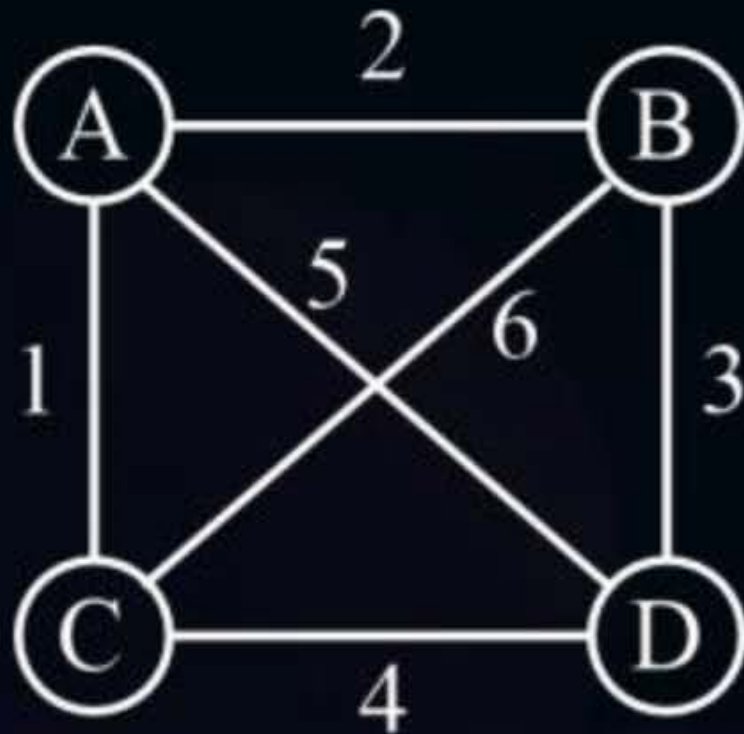




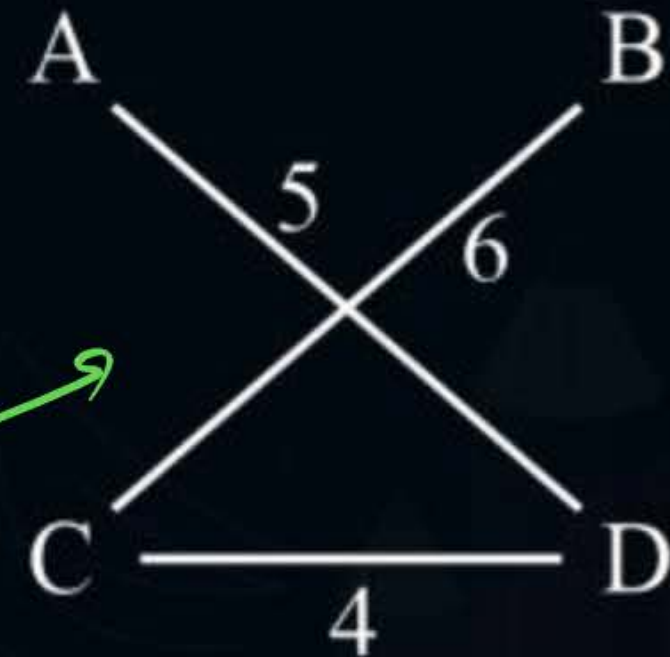
## Topic : Greedy Algorithm



G4:



MCST →



$$\text{Cost} = 6 + 5 + 4 = 15$$

- Not a valid MCST

# [MCQ]

#Q. Consider a complete weighted graph with  $n$  vertices numbered  $V_1$  to  $V_n$ . Two vertices  $V_i$  and  $V_j$  having edge between them has a cost value of  $2|i - j|$ . The weight of minimum cost spanning tree of such a graph is \_\_\_\_\_.

$$n=3, \textcircled{4}$$

**A**  $2n \rightarrow 2 \times 3 = 6 \times$

**B**  $3n - 2 \rightarrow 3 \times 3 - 2 = 9 - 2 = 11 \times$

**C**  $2n - 2$   
 $= 2 \times 3 - 2 = \textcircled{4} \checkmark$

**D**  $2n - 1 \Rightarrow 2 \times 3 - 1$   
 $= 6 - 1 = 5 \times$



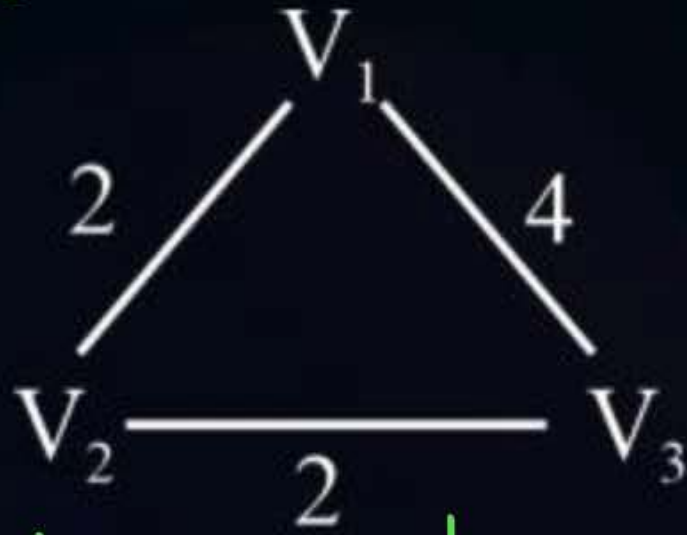


## Topic : Greedy Algorithm



**Solution:**

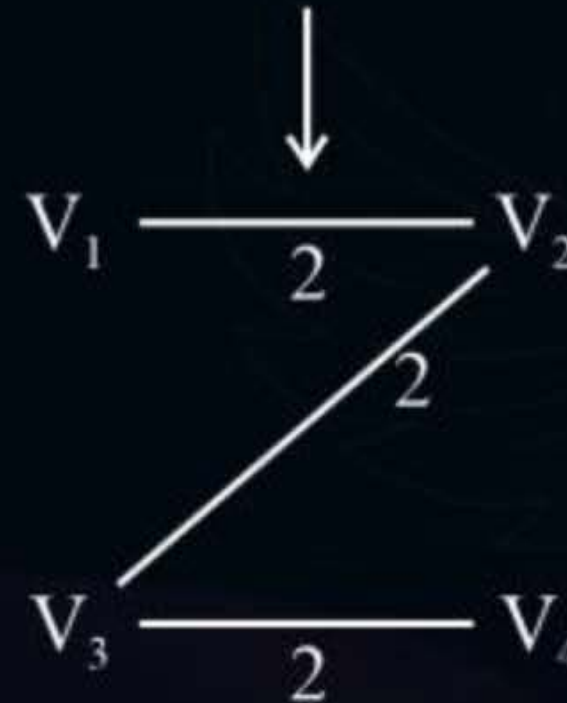
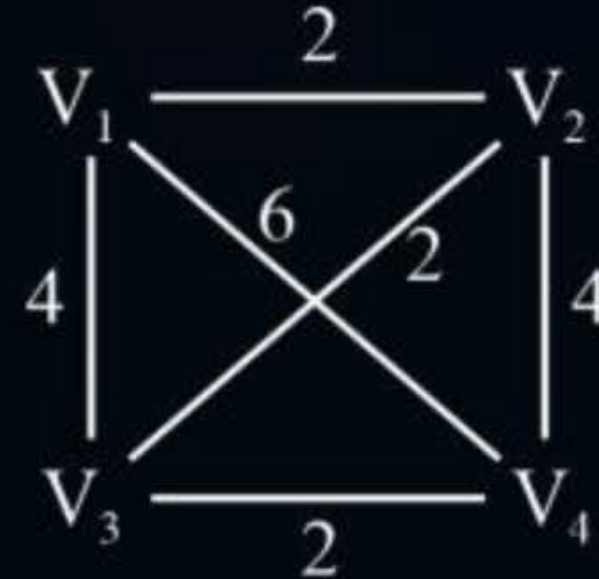
1)  $n=3$ ,  $cost=4$



29



2)



$$V_i - V_{i+1}$$

$$= 2 \times 1$$

$$= \textcircled{2}$$

MST

$$V_1 \overset{2}{-} V_2 \overset{2}{-} V_3 \overset{2}{-} V_4 \overset{2}{-} V_5$$

$$n \Rightarrow e: (n-1)$$

$$\text{cost} = 2 \times (n-1) = \textcircled{2n-2}$$



P48

#Q. Let  $G$  a connected undirected graph of 100 vertices and 300 edges. The weight of a minimum spanning tree of  $G$  is 500. When the weight of each edge of  $G$  is increased by five, the weight of a minimum spanning tree becomes \_\_\_\_\_.

66.6/-

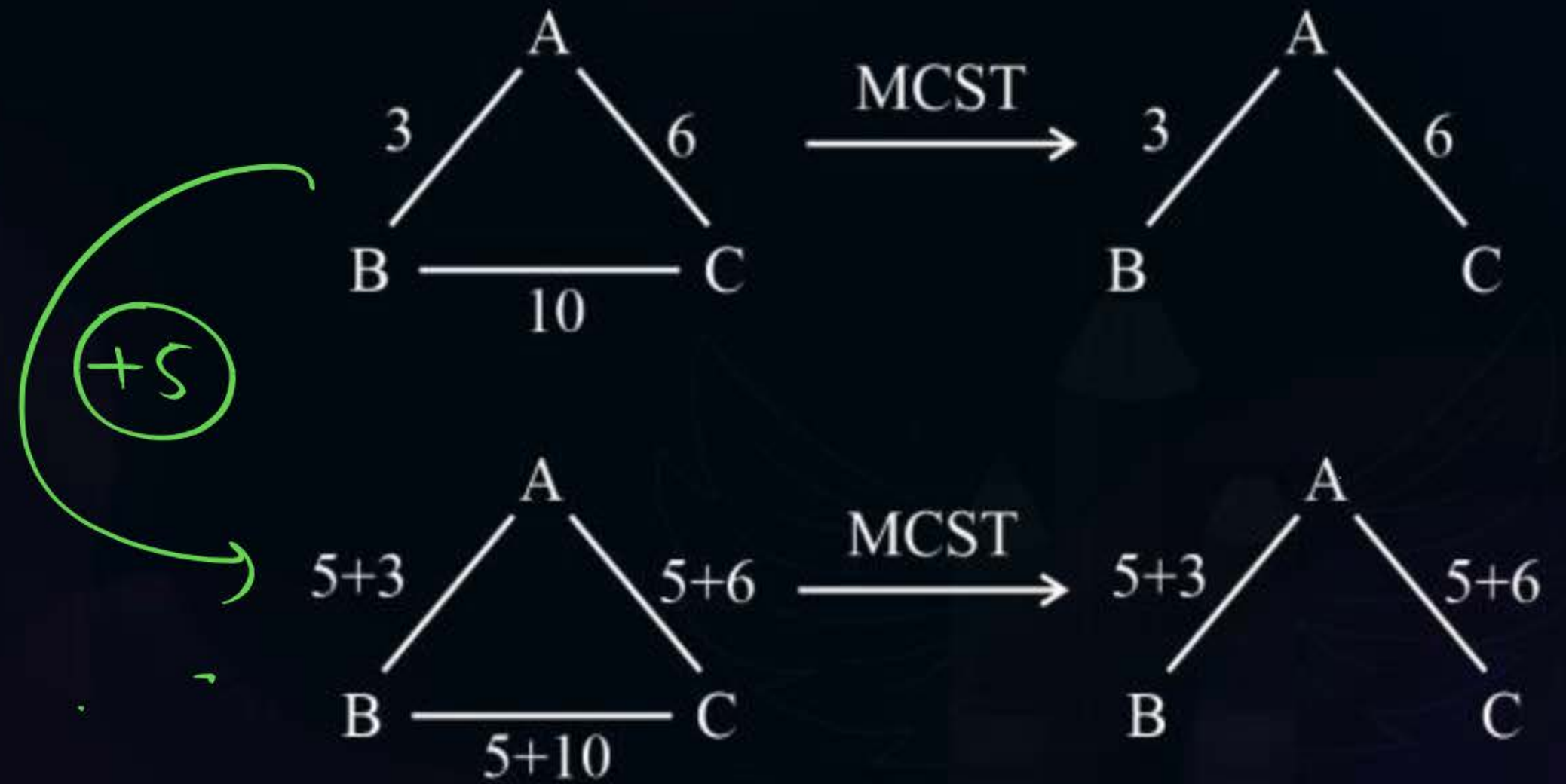
$$\begin{aligned}\text{now } \text{Cost(MST)} &= 500 + 5(n-1) \\ &= 500 + 5 \times (100-1) = 500 + 5 \times 99 \\ &= \boxed{995}\end{aligned}$$



## Topic : Greedy Algorithm



**Solution:**





# [MCQ]

#Q. Consider a graph with 'n' vertices  $n > 2$ . The vertices are numbered  $V_1$  to  $V_n$ . Two vertices  $V_i$  and  $V_j$  are adjacent iff  $0 < |i - j| \leq 2$ . The weight of such an edge is  $i + j$ . The weight of minimum cost spanning tree of such a graph for a value of n is \_\_\_\_\_.

**A**

$$n^2 - n$$

$$n=3, (7) \\ = 9 - 3 = 6 \times$$

**B**

$$n^2 + n + 1$$

$$= 9 + 3 + 1 = 13 \times$$

**C**

$$n^2 - n - 1$$

$$= 9 - 3 - 1 \\ = 5 \times$$

**D**

$$n^2 - n + 1$$

$$= 9 - 3 + 1 = (7) \checkmark$$

$n=4, 16 - 4 + 1 = (13)$

$$70 \sim 6 \%$$

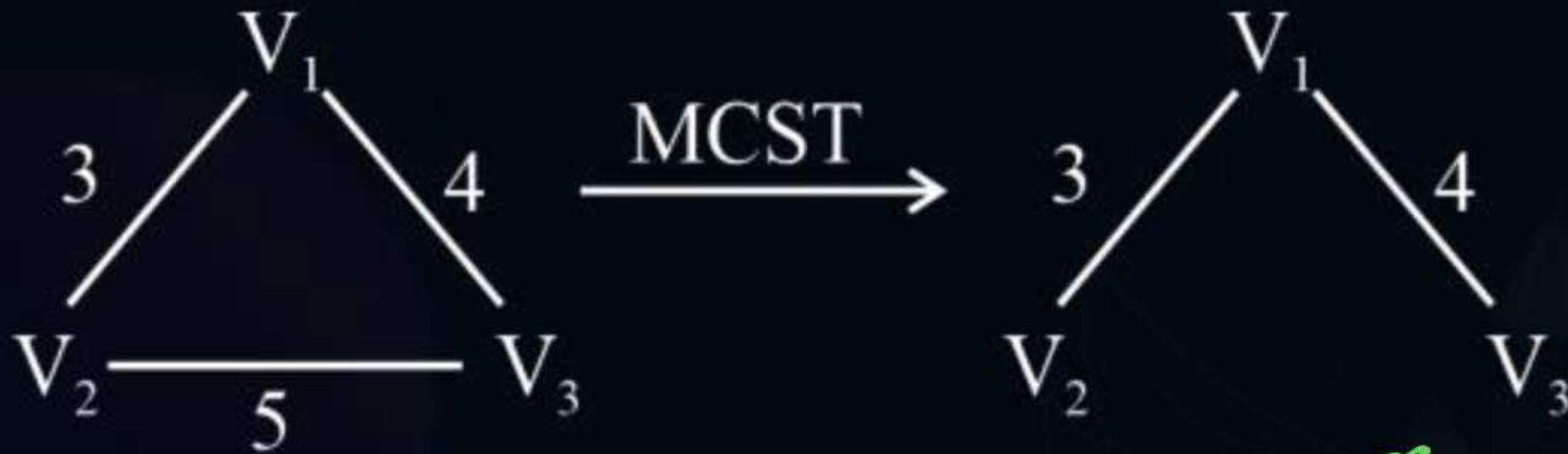


## Topic : Greedy Algorithm



**Solution:**

1)  $n = 3$



$n = 3$

$$\begin{aligned} \text{Cost} &= 3 + 4 \\ &= 7 \end{aligned}$$



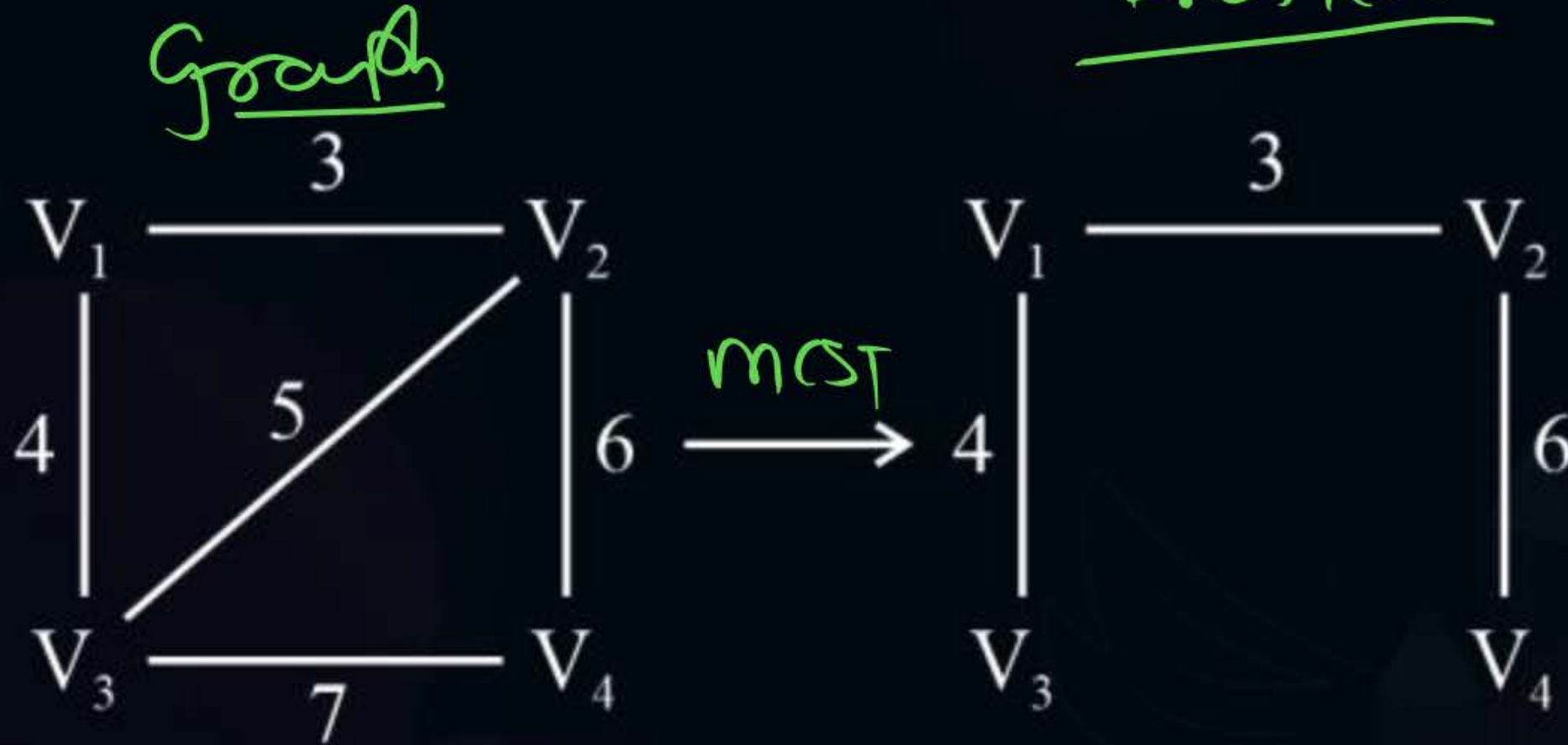




## Topic : Greedy Algorithm



2)  $n = 4$



$$4 + 6 + 3 = 13$$

logically:-

MW

==

$$3+4+6+8 \dots$$

$$= 3+2(2+3 \dots (n-1))$$

~ ~ ~





## 2 mins Summary



Topic

Topic

Topic

Topic

Topic

MCST Construction

PYQs on MCST



**THANK - YOU**